MIL-I-23128B(SH)
23 April 1982
SUPERSEDING
MIL-I-23128A(SHIPS)
24 August 1962
(See 6.5)

MILITARY SPECIFICATION

INSULATION BLANKET, THERMAL, REFRACTORY FIBER, FLEXIBLE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers asbestos-free flexible refractory fiber blanket thermal insulation for control of machinery and equipment at temperatures up to 2000°F (see 6.1).

1.2 Classification. Insulation blankets shall be of the following grades and classes as specified (see 6.2.1):

Grade A - Temperatures up to 1200°F.
Grade B - Temperatures up to 2000°F.

Class 1 - Nominal density, 6 lb/ft^3
Class 2 - Nominal density, 8 lb/ft^3

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specification and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3242, Department of the Navy, Washington, D.C. 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5640

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STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1623 - Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use).

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 167 - Thickness and Density of Blanket-or Batt-Type Thermal Insulating Materials.
C 177 - Steady-State Thermal Transmission Properties by Means of the Guarded Hotplate. (DoD Adopted)
C 201 - Thermal Conductivity of Refractories.
D 1448 - Micronaire Reading of Cotton Fibers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA Traffic Dept; 1616 P Street, NW, Washington, D.C. 20036).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish first article units for first article inspection and approval (see 4.3 and 6.3).

3.2 Material. Insulation shall be composed of asbestos-free refined refractory fibers with or without binders to form a flexible blanket. A certificate to the effect that the material is asbestos-free shall be provided in accordance with the data ordering document included in the contract or purchase order (see 6.2.2).

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Physical properties. Insulation shall meet the physical requirements shown in tables I and II.

TABLE I. Physical requirements.

Characteristic	Grade A	Grade B	Test paragraph
Compressive resistance, to obtain nominal thickness, (max)	50 lb/ft^2	50 lb/ft^2	4.6.2
Tensile strength, (min)	2.5 lb/in^2	Class 1 - 3.0 lb/in^2 Class 2 - 4.5 lb/in^2	4.6.3
Flexibility	No visible rupture		4.6.4
Average fiber diameter, inches by 10^{-4}, (max)	2.5	2.5	4.6.5
Thermal conductivity, BTU-in/sq ft/hr/$^\circ$F (max)	Shall not exceed the values shown in table II		4.6.6
Deterioration of fiber, percent (max)	10	20 percent	4.6.7

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TABLE I. Physical requirements. - Continued

Characteristic	Grade A	Grade B	Test Paragraph
Heat soak aging	24 hours at indicated temperature of:		
	1200°F	2000°F	
Tensile strength, lb/in^2 (min)	2.5	Class 1 - 1.5 lb/in^2 Class 2 - 2.5 lb/in^2	4.6.8 & 4.6.3
Loss in weight, percent (max)	1.5	1.5	4.6.8
Linear shrinkage, percent (max)	1.0	2.5	4.6.8
Fire resistance	Pass	Pass	4.6.9

3.3.1 Thermal conductivity. The maximum values for thermal conductivity shall be in accordance with table II. A certificate of compliance shall be required to show that the material meets the requirements of this specification (see 4.6.6 and 6.2.2).

TABLE II. Maximum thermal conductivity (k) at mean temperature, °F.

Class	300°F	500°F	700°F
1	.40	.55	.70
2	.37	.52	.66

3.4 Dimensions and tolerances. Blanket dimensions shall be as specified (see table III and 6.2.1). A minus tolerance of 1/8 inch in width and length and zero inch for nominal thickness will be permitted. An excess in all dimensions will be permitted.

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TABLE III. Dimensions and tolerances.

Dimension	Feet	Inches	Test paragraph
Thickness		0.5 1.0 1.5 2.0	4.6.10
Width	1.0 2.0 3.0 3.5 4.0 5.0		4.6.11
Length	3.0 4.0 8.0 12.5 24.0 25.0 50.0 100.0		4.6.11

3.4.1 Density. A tolerance value of +20 percent and -15 percent in the nominal densities specified in 1.2 shall be permitted, provided that all other requirements of tables I and III are met (see 4.6.12).

3.5 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is Form OSHA-20 and found as part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4)

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3, 4.4 and 4.6.

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table IV and in 4.6. Certificates of compliance (see 6.2.2) are acceptable for establishing thermal conductivity (see 3.3.1 and 4.6.6) and fire performance (see 3.3 and 4.6.9) (see also 6.3.1 and 6.3.2).

TABLE IV. First article inspection.

Characteristic	Requirement	Test paragraph
Compression	3.3	4.6.2
Tensile strength	3.3	4.6.3
Flexibility	3.3	4.6.4
Fiber diameter	3.3	4.6.5
Thermal conductivity	3.3 and 3.3.1	4.6.6
Deterioration of fiber	3.3	4.6.7
Tensile strength after heat soak	3.3	4.6.8
Loss in weight	3.3	4.6.8
Linear shrinkage	3.3	4.6.8
Fire resistance	3.3	4.6.9
Dimensions and tolerances	3.4	4.6.10 and 4.6.11
Density	3.4.1	4.6.12

4.3.1 First article unit. The first article unit shall consist of one blanket of one thickness from each grade and density. The blanket shall be of sufficient size to permit the examination and tests specified in 4.6.

4.3.2 First article inspection report. The contractor shall furnish a first article inspection report in accordance with the data ordering document included in the contract (see 6.2.2)

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.4.1 and the tests of 4.6, with the exception of thermal conductivity (see 4.6.6) and fire resistance (see 4.6.9).

4.4.1 Visual and dimensional examination. Each of the sample rolls selected in accordance with 4.4.2 shall be surface examined and measured for length, width, thickness, and flexibility to determine conformance with the requirements of this specification.

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4.4.2 Sampling for quality conformance. Samples (see 4.3.1) shall be selected at random from each lot according to table V for the examination and tests specified in 4.4.1 and 4.6 except as noted in section 4.4.2.2.

TABLE V. Sampling.

Number of blankets in inspection lot	Sample number of blankets	Acceptance number (defectives)	Rejection number (defectives)
2 - 8	2	0	1
9 - 15	2	0	1
16 - 25	3	0	1
26 - 50	3	0	1
51 - 90	5	0	1
91 - 150	5	0	1
151 - 280	8	1	2
281 - 500	8	1	2
501 - 1,200	13	1	2
1,201 - 3,200	13	1	2
3,201 - 10,000	20	2	3

4.4.2.1 Test results and consequences. Any roll in the sample (see 4.4.2) containing one or more defects, or deviations from the requirements of this specification shall not be offered for delivery. Additionally, if the number of defective rolls in any sample exceeds the acceptance number for the sample (see table V), the lot represented by the sample shall be rejected.

4.4.2.2 Special sampling. Samples obtained for the tests specified in 4.6.2, 4.6.3, 4.6.5, 4.6.7, 4.6.8 and 4.6.12 shall be selected from each lot. Five rolls shall be chosen at random and one sample from each of the five rolls shall be selected for tensile strength, compressive resistance, physical changes due to heat soak aging. For fineness of fiber, one cluster shall be taken from each of the five rolls and density. For deterioration of fiber, one sample from 2 of the 5 rolls shall be taken (see 4.3.1).

4.4.2.2.1 Test results and convergence. If any of the sample test result averages are found to be not in conformance with this specification, this shall be cause for rejection of the lot represented by the sample.

4.5 Lot. A lot shall consist of insulation blankets of the same grade, density, and thickness produced under similar conditions, and ready for inspection or shipment at the same time.

4.6 Test procedures.

4.6.1 Conditioning samples. Specimens for tests shall be dried by heating to a temperature between 220°F and 230°F, until the weight becomes constant.

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4.6.2 Compressive resistance. Resistance required to compress the blanket to nominal thickness shall be determined by either a dead weight or by mechanical loading. The weight shall be added or applied to the center of the loading plate, 6 by 6 inches by 1/8 inch thick, having a weight of 1-1/4 lbs/ft^2 and progressively increased until the insulation is compressed to its nominal thickness. The weight added per square foot shall be taken as the compression resistance to obtain nominal thickness.

4.6.3 Tensile strength. Tensile strength of the insulation shall be determined with 12-inch wide and 14-inch long specimens cut from the sample blanket in the roll direction. Specimens shall be suspended vertically and held with two clamping devices positioned 12 inches apart. Clamps shall be formed by bolting together at the ends two 14-inch lengths of 1/2-inch pipe. Sponge rubber, 1/4-inch thick, shall be cemented to the outer surface of the pipe in the clamping area to prevent cutting of the specimens. Clamped specimens attached to the jaws of a tensile testing machine, the lower jaw of which moves at a uniform rate of 1 inch plus or minus 0.1 inch per minute shall be pulled to rupture. Tensile strength shall be taken as the average of five determinations and shall be expressed in lb/in^2 of cross sectional area, using the nominal thickness in the computation.

4.6.4 Flexibility. Specimens of insulation (12 inches by 12 inches), shall be bent over an angle of 90 degrees and examined for rupture on the outside surface. Any minor surface separations shall be disregarded. The radii over which the material shall be bent, shall be as follows:

Grade	Radius (inches)
A	2
B	6

4.6.5 Fineness of fiber. Diameter of the fiber shall be determined by either of the following methods:

(a) Microscopic. Diameter of individual fibers shall be determined microscopically on the basis of measuring ten consecutive fibers from each of five clusters selected at random from a sample of insulation. The average of all measurements shall be reported as the fiber diameter.

(b) Air flow. The air flow method as measured by the micronaire instrument in accordance with ASTM D 1448. The micronaire unit shall be calibrated for refractory fibers.

In the event of dispute, the microscopic method shall be the referee method (see 3.3).

4.6.6 Thermal conductivity. Thermal conductivity shall be determined in accordance with ASTM C 177 or ASTM C 201 Annex A1 (see table II). Thermal conductivity tests will be conducted every 12 months, or more often, if found in subsequent examinations that the blanket does not comply with the requirements of tables I and II (see 6.3.1 and 6.3.2).

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4.6.7 Deterioration of fibers. Ten-gram (g) specimens composed of 5/8-inch diameter plugs cut from the insulation, as with a cork borer, shall be used in the fiber deterioration test. Specimens shall be weighed and placed in a covered 8-inch diameter US Standard 25-mesh sieve over a retaining pan secured in a standard laboratory testing sieve on a shaker operated with a horizontal eccentric circular motion at 285 ± 10 revolutions per minute (r/min), and a tapping action of 150 ± 5 strokes per minute. After shaking for 10 minutes with the tapper in operation, material collected in the retaining pan shall be weighed, and the percentage of the total sample calculated. The difference between the values obtained with samples of the insulation in the as-received condition, and those after the heat soak test (see 4.6.8) shall be used to express the percent of fiber deterioration of the insulation as specified in table I.

4.6.8 Physical changes under heat soak. Specimens shall be weighed and measured. Specimens shall then be placed in an electric oven/test furnace and subjected to the maximum temperature for the respective grade for 24 hours. Specimens shall be removed from the oven, reweighed, remeasured and retested for tensile strength. The difference between values obtained with samples of the insulation in the dried condition and those after the heat soak test shall be used to express the loss in weight, and linear shrinkage specified in table I. Samples removed from the oven/furnace shall also be tested for tensile strength as in 4.6.3.

4.6.8.1 Tensile strength (see 4.6.3).

4.6.9 Fire resistance. Each density of insulation blanket shall be fire tested in accordance with MIL-STD-1623 for first article inspection only. Fire resistance tests will be conducted every three years, or more often, if found in subsequent examination that the blanket does not comply with the requirements of table I (see 6.3.1).

4.6.10 Thickness. Actual thickness shall be determined in accordance with the method specified in ASTM C 167 (see 3.4).

4.6.11 Length and width. Nominal length and width shall be determined by direct measurement, using a standard measuring instrument with 1/16-inch graduations (see 3.4).

4.6.12 Density. Density shall be determined in accordance with the method specified in ASTM C 167, except that the nominal thickness as determined in 4.4.2 shall be used to calculate the density (see 3.4.1).

4.7 Packaging inspection. Packing and marking shall be examined for conformance with the requirements of section 5.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packing. Packing shall be level A, B or C, as specified (see 6.2.1).

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5.1.1 Level A. Insulation blankets shall be packed in containers conforming to PPP-B-636, class weather resistant, except that limitations on inside dimensions of box shall not apply. Containers shall be closed, waterproofed, and reinforced with tape in accordance with method V of the appendix to PPP-B-636. Weight limitations of the specification shall not be exceeded.

5.1.2 Level B. Insulation blankets shall be packed in fiberboard boxes conforming to class domestic of PPP-B-636; weight limitations of the specification shall not be exceeded. Box closures shall conform to method I, as specified in the appendix to PPP-B-636.

5.1.3 Level C. Insulation blankets shall be packed in containers in a manner which will insure acceptance by common carrier at the lowest rates, and will afford protection against physical damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general shall conform to the Uniform Freight or National Motor Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation. The contractor's normal retail or wholesale packing methods may be utilized when such meet the requirements of this level.

5.2 Marking. In addition to any special marking required by the contract (see 6.2.1), exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Grade A. Grade A material is intended as a lower-temperature outer layer component for Grade B material, or as insulation where service temperatures do not exceed 1200°F.

6.1.2 Grade B. Grade B material is intended for use as thermal insulation of machinery and equipment such as steam or gas turbines, boilers, and piping operating temperatures up to 2000°F.

6.1.3 Classes. Thermal conductivity decreases with increasing density of the insulation. Class 2 insulation is superior and stronger than class 1, and should be specified if increased weight is not significant for the intended application.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grade and class required (see 1.2).
(c) When a first article unit is required for inspection and approval (see 3.1, 4.3.1, and 6.3).
(d) Thickness, width, and length required (see 3.4).
(e) Level of packing required (see 5.1).
(f) Special marking if required (see 5.2).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
3.2	Certificate of compliance	DI-E-2121	----
3.3.1	Certificate of compliance	DI-E-2121	----
4.3.2	First article inspection report	UDI-T-23790	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Test report data. The contract will specify that quality conformance inspection (see 4.4) and first article test reports shall be prepared by the manufacturer and sent to both the contracting activity and Department of the Navy, Naval Sea Systems Command, Washington, DC 20362, Attn: NAVSEA Code 05E3.

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 Guidelines. The following guidelines should be used to determine the necessity for required first article inspection for fire resistance and thermal conductivity:

(a) If within the stated time period (see 6.3.2) preceding the date of invitation for bids the blanket has not been tested by an acceptable testing laboratory and found in compliance with the requirements of this specification, or

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(b) If the material offered for delivery is not manufactured the same in all respects as that previously tested.

6.3.2 Time period. The maximum time interval between valid tests for fire resistance and thermal conductivity are as follows:

(a) Fire resistance - 36 months, except as noted in 6.3.1(b).
(b) Thermal conductivity - 12 months, except as noted in 6.3.1 (b).

6.4 Comercial sizes. Sizes and thicknesses of insulation covered by this specification are industry standard. Other sizes and thicknesses may be obtained upon request from the contracting activity.

6.5 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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